

1. A Convolutional Layer vs. A Fully Connected Layer:

A. connectivity & weights:

⇒ a convolutional layer is **sparsely connected** and utilizes **weight sharing** with the same feature map.

⇒ a fully connected layer is **fully connected**, with entirely **different weights** for each individual neuron.

B. num of unique weights:

$$\Rightarrow UW_c = (K \times K \times C_{in} + 1) \times C_{out}, \left[\begin{array}{l} UW_c = \text{num of unique weights in a convolutional layer} \\ K \times K \times C_{in} = \text{size of a convolutional kernel.} \\ K = \text{the side length of a convolutional kernel as seen in a 2D plane} \\ C_{in} = \text{num of input channels in a convolutional layer.} \\ / = \text{num of bias weight for a single neuron in a convolutional layer} \\ C_{out} = \text{num of output channels in a convolutional layer.} \end{array} \right]$$

$$= W_c \times C_{out}, \left[\begin{array}{l} W_c = \text{num of weights for a single neuron in a convolutional layer.} \\ C_{out} = \text{num of output channels in a convolutional layer.} \end{array} \right]$$

$$\Rightarrow UW_{fc} = W_{fc} \times N_{fc}, \left[\begin{array}{l} UW_{fc} = \text{num of unique weights in a fully connected layer} \\ W_{fc} = \text{num of weights for a single neuron in a fully connected layer} \\ N_{fc} = \text{num of neuron in a fully connected layer} \end{array} \right]$$

C. num of MACs (Multiply-Accumulate Operations):

$$\Rightarrow MAC_{sc} = (K \times K \times C_{in}) \times (\text{Width} \times \text{Height} \times C_{out}), \left[\begin{array}{l} MAC_{sc} = \text{num of MACs in a convolutional layer} \\ K \times K \times C_{in} = \text{size of a convolutional kernel.} \\ K = \text{the side length of a convolutional kernel as seen in a 2D plane} \\ C_{in} = \text{num of input channels in a convolutional layer.} \\ \text{Width} \times \text{Height} = \text{the grid dimensions of the neurons arranged within a single feature map in a convolutional layer} \\ C_{out} = \text{num of output channels in a convolutional layer.} \end{array} \right]$$

$$= DI_c \times N_c, \left[\begin{array}{l} DI_c = \text{num of data inputs for a single neuron in a convolutional layer.} \\ N_c = \text{num of neurons in a convolutional layer.} \end{array} \right]$$

$$\Rightarrow MAC_{sc} = DI_{fc} \times N_{fc}, \left[\begin{array}{l} MAC_{sc} = \text{num of MACs in a fully connected layer.} \\ DI_{fc} = \text{num of data inputs for a single neuron in a fully connected layer.} \\ N_{fc} = \text{num of neurons in a fully connected layer.} \end{array} \right]$$

D. num of FLOPs (Floating Point Operations):

$$\Rightarrow FLOPs_c = 2 \times MAC_{sc} + N_c, \left[\begin{array}{l} FLOPs_c = \text{num of FLOPs in a convolutional layer} \\ MAC_{sc} = \text{num of MACs in a convolutional layer} \\ N_c = \text{num of neurons in a convolutional layer.} \end{array} \right]$$

$$\Rightarrow FLOPs_{fc} = 2 \times MAC_{sc} + N_{fc}, \left[\begin{array}{l} FLOPs_{fc} = \text{num of FLOPs in a fully connected layer.} \\ MAC_{sc} = \text{num of MACs in a fully connected layer.} \\ N_{fc} = \text{num of neuron in a fully connected layer} \end{array} \right]$$

2. num of weights for a single neuron in a fully connected layer:

$$W_{fc} = (\text{num of neurons in the previous convolutional layer}) + 1, \left[\begin{array}{l} W_{fc} = \text{num of weights for a single neuron in a fully connected layer} \\ / = \text{num of bias weight for a single neuron in a fully connected layer} \end{array} \right]$$

3. num of neuron in a fully connected layer:

$$N_{fc} = \text{num of classes defined in a classification issue}, [N_{fc} = \text{num of neuron in a fully connected layer}]$$

4. num of data inputs for a single neuron in a fully connected layer:

$$DI_{fc} = \text{num of neurons in the previous convolutional layer}, [DI_{fc} = \text{num of data inputs for a single neuron in a fully connected layer}]$$